Penalized Portfolio Optimization

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Abstract

This research reviews practical considerations of long-short portfolio optimization problems. It studies a few penalty forms to address the limitation of pure constraints.

1 Background of Portfolio Construction

1.1 Principles of Portfolio Construction

Modern portfolio construction involves optimizing an objective function defined over a vector of n security positions $w \in \mathbb{R}^n$, where positive (negative) values represent long (short) positions. An example of a classic objective function is Markowitz's Mean-Variance framework:

$$\min_w \left[\frac{\lambda}{2} w' \Sigma w - r' w \right], \tag{1}$$

where r is a vector of expected returns, λ is an exogenous risk-aversion parameter, and Σ is a covariance matrix of asset returns. The design is driven by the desire to strike a balance between risks and payoffs.

Objective functions can take other forms, such as L_2 norm of the deviation from a weight vector $\tilde{w}$. A given vector $\tilde{w}$ can be a benchmark portfolio to track, or a target portfolio driven by some forecasting models. It is often desirable to mimic such a portfolio as closely as possible. One reasonable approach is to minimize this objective:

$$\min_w \|w - \tilde{w}\|_2^2.$$

A non-negative diagonal weighting scheme matrix Γ can also be introduced to generalize the above objective function further,

$$\min_w \left[\frac{1}{2} (w - \tilde{w})' \Gamma (w - \tilde{w}) \right] \equiv \min_w \left[\frac{1}{2} w' \Gamma w - \tilde{w}' \Gamma w \right].$$

This problem will be elaborated upon in section 5.1.

A majority of portfolio construction problems can be formulated in certain quadratic forms like the above. Typical constraints include position limits, and total portfolio turnovers, etc.

Appropriate solution methods is contingent on the problem's structure. Objective (1) is convex if Σ is positive semi-definite and if its constraints maintain a convex feasible set. The L_2 norm objective acts similarly, placing both objectives within the reach of Disciplined Convex Programming (DCP). The convexity makes any local optimum a global minimum, and the Karush-Kuhn Tucker (KKT) conditions provide a complete characterization of optimality which we will elaborate upon later in section 2.3.

The widespread efficiency and availability of DCP solvers means convex problems can be optimized notably quickly.

2 Implementation Framework

Let us focus on a family of problems with a base quadratic objective component $q(w)$, which captures the conventional Mean-Variance problem and target tracking problem.

Constraints can model numerous types of boundaries, including risk modeling. This includes factor exposure bounds, sector or industry neutrality, and tracking error limits.

Inputs such as return estimates, risk model coefficients, and target forecasts in $\tilde{w}$ are all estimates subject to statistical errors. Thus, a problem subject to overly permissive constraints is susceptible to concentrated exposure on assets with potentially poor signals. Constraints like turnover and position limits act as a form of regularization against over-adjusting to noisy models.

On the other hand, we look to avoid overly-tight constraints. While constraints have utility, each imposition reduces the feasible region of potential solutions. An excessively rigid set of boundaries can force the optimizer to a suboptimal corner of a feasible set that reflects very little of the objective. Therefore, we shall try to set constraints such that they are reflective of practical and risk limits rather than as the determining factor in calculating an optimal point. In other words, we do not wish for constraints to "drive" the solution; the solver should be dictated in part by $\tilde{w}$.

While potential payoffs are often uncertain, transaction costs are incurred upon execution. Turnover budgets and individual turnovers are imposed, so a portfolio is discouraged from chasing marginal returns for excessive transaction costs.

Regardless of the form of a constraint, we show a preference for preserving convexity of the overall feasible region. Non-convexity in a feasible region makes it far more difficult for a solver to reliably calculate an optimal solution, often trapping the algorithm in local minima. Linear equalities, inequalities, as well as second-order cone constraints define half-spaces, hyperplanes, and second-order cones respectively. These all define convex sets. For constraints that are non-smooth but inherently convex (such as absolute values or piecewise functions), we exercise exact epigraph reformulations with auxiliary variables. For constraints that are truly non-convex, we employ convex relaxations to approximate the feasible space while maintaining computational tractability.

2.1 Constraints

Subjecting w to minimum and maximum individual asset boundaries can be accomplished with strictly linear constraints. Constraints are defined by vectors $\underline{w}$ and $\overline{w}$, respectively.

$$\underline{w} \leq w \leq \overline{w}.$$

Similarly, limiting each individual asset's turnover to values denoted by vector t can be accomplished through more linear inequalities.

$$|w - w^*| \leq t.$$

Net exposure ρ is equal to total long exposure minus total short exposure. Essentially, it equals $e'w$ where $e \in \mathbb{R}^n$ is vector with all ones. The net exposure takes the form

$$|e'w| \leq \rho.$$

There are several ways to impose constraints on characteristic exposures. One of the ways is to express n assets' exposure in m attributes utilizing an $m \times n$ matrix $\mathbf{S}$. For example, a potential $\mathbf{S}$ for 3 assets across energy and financial attributes is shown in Table 1.

Table 1: Example Factor Loading Matrix.

	Asset A	**Asset B**	**Asset C**
Financial	0.6	0.3	0.5
Energy	0.4	0.7	0.5

Given a factor loading matrix $\mathbf{S}_{m\times n}$, the total portfolio exposures to the m factors are $\mathbf{S}w$. The exposure limits can be represented as

$$\underline{s} \leq \mathbf{S}w \leq \overline{s} \tag{2}$$

Total turnover bounds aggregate change in position across all assets using an L_1 constraint on the difference between w and w^*, bounded by total turnover T.

$$\|w^* - w\|_1 \leq T \tag{3}$$

This is meaningfully different from the individual turnover constraint, as it balances a total budget, instead of individual change. In practice, individual turnover is often set to be less restrictive (such that many assets in w are not equal to their maximum turnover) than total turnover, which is often pushed to T. If individual turnover constraints are too restrictive, then a maximum turnover is essentially implied. However, it often leads to less diverse portfolio compositions, as variation between assets is diminished.

The absolute value summation constraint $\sum_{i=1}^{n} |w_i - w_i^*| \leq T$ is not natively solvable in many solvers. Although the constraint is convex, it is not affine, and is not differentiable at $w_i = w_i^*$. To alleviate this, slack variables are created to reformulate the constraint. This operation is explained in detail in section 2.2.

Gross exposure, denoted by G, is another value commonly used in portfolio optimization. It is calculated explicitly through

$$\|w\|_1 = G. \tag{4}$$

(4) is typically an equality constraint, as firms often wish to keep a specific amount of capital in a portfolio. However, creating this equality constraint explicitly results in non-convexity (to be proved later). This is different from (3). This constraint is simply unviable in its current form, while (3) is convex and can be reformulated in different expressions. Thus, we will need to find a way to reformulate or avoid this non-convexity. The convexity of (4) is addressed in section 2.4.

It's important to note that in long-only or short-only portfolios, non-convexity is not an issue as (4) can be expressed without absolute values (since all w_i have the same signs). However, in long-short portfolios, an L_1 norm is required.

2.2 Treatment of Total Turnover

Numerous solvers do not natively accept absolute value constraint expressions, making it not directly compatible with (3). A slack variable operation is created to circumvent this. A buffer vector $y \in \mathbb{R}^n$ with the same size as w is subjected to the constraint

$$\begin{aligned} |w^* - w| &\leq y, \\ e'y &\leq T, \end{aligned}$$

where y is only bounded below by $|w - w^*|$. Since increasingly y_i will only make this condition more difficult to satisfy and doesn't affect the objective value, y_i will be forced downwards toward the tightest feasible value $y_i = |w_i^* - w_i|$. In doing so, (3) is replaced with strictly linear expressions. At the optimal value, it must satisfy

$$e'y = \|w^* - w\|_1 \leq T.$$

These reformulated constraints are strictly linear.

2.3 Karush-Kuhn-Tucker and Global Optimality Conditions

Because the portfolio construction problems considered henceforth are convex, their solutions are exact and can be calculated via DCP. Suppose a quadratic objective is bounded by affine constraints that create a convex feasible set. The Karush-Kuhn-Tucker (KKT) conditions [1] define the framework for determining optimality. For a convex objective function with convex inequality and affine equality constraints, a point that satisfies the conditions is guaranteed to be a global minimum. Using a convex quadratic form to satisfy these conditions calls for two conditions: a positive semi-definite Hessian matrix, which indicates convexity, and a constraint qualification, which we will fulfill through using affine constraints.

In its current state, the objective function satisfies these conditions as it contains purely quadratic terms, and almost all constraints implemented thus far satisfy, the exception being gross exposure.

2.4 Difficulties of Gross Exposure

A set $\mathcal{C} \subset \mathbb{R}^n$ is convex if and only if for any two points a and b in $\mathcal{C}$ and every $\theta \in [0, 1]$ the following is also true:

$$\theta a + b(1 - \theta) \in \mathcal{C}. \tag{5}$$

We will prove that for fixed $G > 0$, the set $\mathcal{S} = \{w \in \mathbb{R}^n : \|w\|_1 = G\}$ is not convex. This argument does not restrict itself to properties of the L_1 norm. Rather, it uses two axioms every norm satisfies:

1. Absolute Homogeneity: $\|\alpha w\| = |\alpha|\|w\|$ for any scalar α.
2. Positive definiteness: $\|w\| = 0 \Leftrightarrow w = 0$.

Take any $w \in \mathcal{S}$. By homogeneity with $\alpha = -1$,

$$\|-w\|_1 = |-1| \cdot \|w\|_1 = \|w\|_1 = G,$$

so $-w \in \mathcal{S}$ as well. Now take $\theta = \frac{1}{2}$ and form (5) of w and $-w$:

$$\theta w + (1-\theta)(-w) = 0.$$

By positive definiteness, $\|0\|_1 = 0$. Since $G > 0$ by assumption, $0 \neq G$, so

$$\theta w + (1-\theta)(-w) = 0 \notin \mathcal{S}.$$

We have shown that $\theta w + (1-\theta)(-w) \notin \mathcal{S}$ for both w and $-w$ in $\mathcal{S}$. By (5), the set $\mathcal{S}$ is not convex.

One way to overcome this non-convexity is forcing the sign of w_i to match the sign of $\tilde{w}$ for all i. Let $G^+ > 0$ be the desired long exposure, and $G^- < 0$ be the negative counterpart. The total gross exposure becomes $G^+ - G^- = G$. We then introduce the reformulated gross exposure constraints:

$$\begin{aligned}
\sum_{i \in \mathcal{W}^+} w_i = G^+, \quad & \mathcal{W}^+ = \{i | \tilde{w}_i > 0\}, \\
\sum_{i \in \mathcal{W}^-} w_i = G^-, \quad & \mathcal{W}^- = \{i | \tilde{w}_i < 0\}, \\
\sum_{i \in \mathcal{W}^0} w_i = 0, \qquad & \mathcal{W}^0 = \{i | \tilde{w}_i = 0\}.
\end{aligned}$$

These linear constraints avoid the non-convexity presented by (4). Furthermore, it also presents certain practicalities. Obviously, the shrinking feasible set will not guarantee the global optimal solution. Even Mixed-Integer Convex Programming (MICP) has difficulty finding a global optimal solution in a timely and resource-efficient manner.

Other local search methods may improve upon this approach:

- Solve the original problem with inequality, and use the signs of the solution vector to derive $\mathcal{W}^+$, $\mathcal{W}^-$, and $\mathcal{W}^0$.
- After one iteration, flip the signs of whose securities $|w_i|$ is within a small band ϵ to update $\mathcal{W}^+$, $\mathcal{W}^-$, and $\mathcal{W}^0$. Solve again. If there is improvement, increase ϵ and repeat. If not, stop.

- Solve the original problem with inequality, and scale all the weights such that $\|w\| = G$. Feed the results back to solve again.

3 Soft Cap of Transaction Costs

Hard constraints that bound turnover and other portfolio characteristics have certain downsides. Violating a hard constraint by a negligible amount and by a comparatively huge amount have the exact same effect, meaning rounding errors, tiny violations, and genuine infeasibility are treated the same. Furthermore, if for some i, $|w_i^* - \tilde{w}_i|$ is greater than t_i, there is a sizable chance that the stock should be considerably skewed in the direction of $\tilde{w}_i$. In those cases, it is desirable to permit certain violations of hard constraints. To do this, we introduce soft constraints.

Soft constraints allow such violations while discouraging large ones by adding a penalty to the objective. We will consider three distinct types of soft constraints: an alpha penalty, an Elastic Net, and a Berhu penalty.

3.1 Soft Cap of Total Turnover

The trading from w^* to w incurs transaction costs. Naturally, this requires balancing objective and turnover to avoid paying deterministic upfront costs on insignificant or uncertain transactions. Imposing a hard budge limit on total turnover, however, handicaps the portfolio construction on achieving its objective.

In addition to a hard upper bound T on total turnover $\|w - w^*\|_1$ or as its replacement, we can introduce a slack variable $z > 0$ to relax the hard bound such that

$$\|w - w^*\|_1 \leq \tilde{T} + z,$$

and add a penalty term to the objective function $q(w)$,

$$p(z) = \alpha z^2.$$

The exogenous parameter $\alpha > 0$ determines the cost of going above $\tilde{T}$. The variable z will be optimized along with w. The quadratic term has the benefit of having a comparable magnitude with many objective function forms of $q(w)$ such that the choice of α can be more intuitive. It is completely reasonable to replace it with $\alpha|z|$. The presence of an absolute value poses no difficult here given that $z \geq 0$.

This formulation is very intuitive. The total turnover is allowed to exceed $\tilde{T}$, but pays a penalty to do so. If turnover is less than $\tilde{T}$, no penalty is added. The larger the value of α, the more the final turnover is discouraged from exceeding T, and vice versa. This "soft cap" is not a rigid hard bound, allowing some room to trade more when situation dictates that. When both a "hard cap" and a "soft cap" are present, we generally have $\tilde{T} < T$ such that T serves as a final safeguard on the total turnover.

4 Regularization of Noisy Trades

While it is a common practice to cap the total turnover $\|w-w^*\|_1$ or limit some weighted transaction costs $e'\Gamma|w-w^*|$ by a budget, it is also desirable to reduce the amount of inconsequential small trades. They are either results of some computational or statistical noises, or are simply too insignificant to justify the execution or handling costs.

We wish to induce a degree of sparsity in a portfolio's turnovers, eliminating noisy trades. A common approach to achieve it involves setting a certain threshold for $|w_i^* - w_i|$ below which those $\{w_i\}$ will be forced to stay at $\{w_i^*\}$, meaning no trade allowed. However, this hard rule may be too rigid, and the threshold setting also seems rather arbitrary.

This problem is very similar to variable selection in various regression frameworks where coefficients of some variables are shrunk towards zero to yield more parsimonious models. It is thus not surprising that similar techniques can be employed to regularize $w - w^*$.

4.1 Elastic Net

Ridge Regression [5] softens rigid parameters through adding an L_2 norm scaled by a multiplier. This allows a soft constraint while simultaneously discouraging it past a certain point. Lasso regression [11] modifies the L_2 norm to instead be an L_1 norm. The Elastic Net [12] uses both Lasso and Ridge simultaneously. This concept can be applied to portfolio optimization, penalizing noise trades:

$$\beta\|w - w^*\|_1 + \frac{1-\beta}{2}\|w - w^*\|_2^2, \tag{6}$$

where $\beta \geq 0$ and $\lambda > 0$ are adjustable exogenous parameters. The L_1 penalty promotes sparsity. It is a mechanism that drives small weight turnovers to 0, reducing the number of assets to trade. The L_1 multiplier β governs the degree of sparsity on the portfolio. A larger β reflects a higher marginal cost of initiating a new trade. This shrinks the amount of sensible trades, forcing the optimizer to only execute trades where expected returns outweigh a fixed transaction cost. The parameter λ controls the overall scale of this penalty term relative to the objective function $q(w)$.

The L_2 penalty promotes smoothness. It prevents a single element from changing drastically. Hence, turnover among correlated assets is distributed more evenly than with the absolute value penalty. For example, the L_2 penalty resulting from two turnovers 0.2 and 0.2 is smaller than the penalty resulting from 0.1 and 0.3, even though the total turnover is the same. Thus, the L_2 turnover results in spreading turnover more evenly across assets. The β parameters governs relative strength of penalties between the two terms.

A penalty term (6) and an alpha gating quadratic term can be added to a base quadratic objective $q(w)$:

$$n(w) = \lambda\left(\beta\|w - w^*\|_1 + \frac{1-\beta}{2}\|w - w^*\|_2^2\right),$$

where $\lambda > 0$ determines the relative strength of this penalty term in relation to the the objective function $q(w)$ and other penalty terms. The greater λ is the more trades will be pushed towards

zero.

Each term in the objective is a L_1 norm, a L_2 norm, or a quadratic function. Since both β and λ are non-negative, the entire objective function is strictly convex. This guarantees that any local optimal solution is also the global optimum. Note that an L_1 norm in the objective function is permissible, while an L_1 norm equality constraint is not convex.

4.2 Berhu's (Reverse Huber's) Loss

Elastic Net penalties subject portfolio weights to both L_1 and L_2 penalties, regardless of the magnitude of an asset's turnover. We can introduce a piecewise function that contains separate L_1 and L_2 penalties, then set a specific threshold δ where one or the other is used. To do this, we utilize a Berhu's (Reverse Huber) Loss.

Frequently used in regressions, the Huber's Loss function takes the form

$$h(u_i) = \begin{cases} \frac{u_i^2}{2\delta} & \text{if } u_i \leq \delta, \\ u_i - \frac{\delta}{2} & \text{if } u_i > \delta. \end{cases} \tag{7}$$

It applies a quadratic penalty for small deviations, and a linear penalty for large deviations. This mitigates the influence of extreme outliers in regression while maintaining differentiability throughout.

Berhu's Loss, also known as Reverse Huber's Loss, inverts the order of the quadratic and linear functions, and is added to the objective function; for a turnover u_i and a threshold scalar $\delta > 0$, the Berhu Loss is defined as

$$b(w_i) = \begin{cases} u_i & \text{if } u_i \leq \delta, \\ \frac{u_i^2+\delta^2}{2\delta} & \text{if } u_i > \delta. \end{cases} \tag{8}$$

This function is similarly to Elastic Net, but only applies either an L_1 norm or L_2 norm at a time. For minor weight adjustments where $u_i \leq \delta$, the penalty is linear, promoting sparsity. For significant weight adjustments where $u_i \geq \delta$, the penalty becomes quadratic and prevents concentrated trades.

Intuitively, Elastic Net sharpens but does not completely eliminate the smooth edges of the feasible set while the Berhu's function shrinks the areas of pointy edges but still preserve the indifferentiable areas. Both encourage more control variables to go towards zero upon contact with the objective function at the optimal solution, just in slightly different fashions. As a trade increases beyond δ, Berhu's function becomes quadratic while a comparable Elastic Net still maintains a linear term. Therefore, Berhu's functions penalize the large outliers more aggressively than the similar Elastic Net functions.

This function is convex and continuous everywhere. That said, its piecewise nature makes it difficult to implement directly into a solver, as many solvers do not accept piecewise functions natively or have a designated Berhu function. To resolve this, we'll create an epigraph formulation with auxiliary variables.

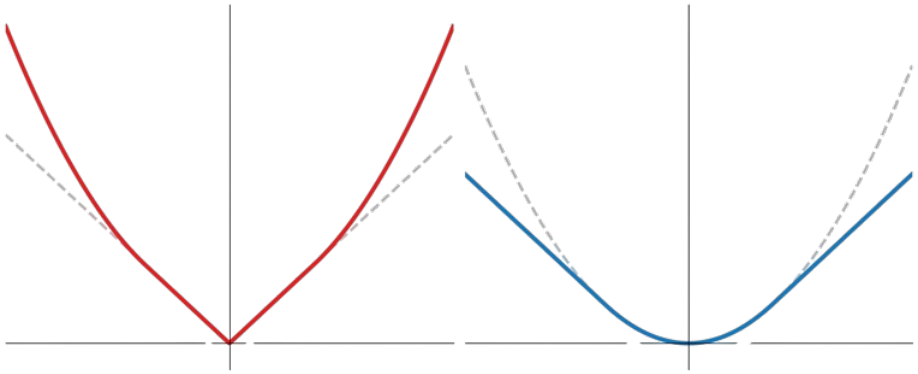


Figure 1: Berhu's Loss (Left) and Huber's Loss (Right).

Define two non-negative auxiliary variables $s_i \geq 0$ and $t_i \geq 0$ subject to the following constraints:

$$s_i + t_i \geq u_i, \quad s_i \leq \delta.$$

The Berhu Loss then becomes

$$b(w) = \sum_i \left(s_i + t_i + \frac{t_i^2}{2\delta} \right), \tag{9}$$

We can prove equivalence with basic casework. In the first case, $u_i \leq \delta$. To minimize the penalty, the solver will set $t_i = 0$, since the quadratic penalty term $\frac{t_i^2}{2\delta}$ grows more rapidly than the linear term s_i, being more "expensive" in the objective function. The constraint $s_i + t_i \geq u_i$ forces $s_i = u_i$, making the penalty exactly u_i, matching the linear branch of (8).

In case two, $u_i > \delta$. The minimizer pushes s_i to its upper bound δ, resulting in t_i being non-zero:

$$t_i = u_i - \delta,$$

while the cost becomes

$$\delta + (u_i - \delta) + \frac{(u_i - \delta)^2}{2\delta} = \frac{u_i^2 + \delta^2}{2\delta}.$$

This matches the value of the quadratic branch of (8). Thus, we've reformulated the piecewise Berhu function (8) into only an epigraph with auxiliary variables.

Let $y = s + t$ and notice $|w - w^*| \leq y$, where $|s| \leq \delta$ and $t \geq 0$. A penalty term based on (9) can then be defined as

$$b_t(|w - w^*|) = \gamma \sum \left(s_i + t_i + \frac{t_i^2}{2\delta} \right) = \gamma \sum_i \left[u_i + \frac{1}{2\delta} \max[u_i - \delta, 0]^2 \right], \tag{10}$$

where $\gamma > 0$ is an adjustable exogenous scale coefficient that determines the scale of this term relative to the objective function $q(w)$ and other penalty terms present. All terms in the objective function are either quadratic or linear, so the function is convex.

Once again, it is important to consider the impact of changes in the parameters. Threshold δ

is the inflection point where the penalty transitions from linear to quadratic. A small δ means the optimizer switches to quadratic penalties at very small turnover values, resulting in a more dense portfolio with fewer turnovers pushed to exactly 0.

A large δ means the penalty remains linear for a wider range of turnovers, promoting sparsity in the same way the Elastic Net acted.

As stated before, the difference between (6) and (9) is that (9) imposes only one penalty per asset, while (6) imposes both simultaneously. So in an Elastic Net, the "effective" L_1 penalty is always slightly increased, even when u_i is almost negligible. For the Berhu's Loss, one can set a concrete threshold δ past which they wish to start imposing quadratic penalties.

5 Huber's Loss Objective Function

5.1 Target Tracking

A target vector $\tilde{w} \in \mathbb{R}^n$ may represent a benchmark portfolio or a model driven portfolio. It is often desirable to form portfolios as close as possible to such a given portfolio subject to various constraints.

$\tilde{w}$ is normalized such that it possesses a consistent scale and is comparable across adjustments. In long-short portfolios, this typically takes the form of a gross exposure normalization:

$$\|\tilde{w}\|_1 = G.$$

Normalizing fixes scales and units in which the objective function and subsequent constraints are expressed, and avoids degenerate solutions where magnitudes are dilated rather than truly altering relative composition of the portfolio.

Gross exposure constraints will also be placed upon w, the new portfolio being optimized. Exact gross exposure constraints that apply to w and $\tilde{w}$ ensure equal scale. This gross exposure constraint must be an equality to be effective, and is discussed in section 2.4.

Up to this point, the base objective family $q(w)$ we used was a quadratic function along with some linear or quadratic penalty terms represented by $p(w)$, an example of which is

$$q(w) = \|\tilde{w} - w\|_2^2 + p(w).$$

Many solvers can handle such problems very efficiently.

The previously stated Huber's Loss function (7) can be used as a completely different objective function, not just a penalty.

For a tracking error vector $v = w - \tilde{w}$, and a threshold parameter δ, the piecewise Huber's Loss function is defined as

$$h(v_i) = \begin{cases} \dfrac{v_i^2}{2\delta} & \text{if } v_i \leq \delta, \\ v_i - \dfrac{\delta}{2} & \text{if } v_i > \delta. \end{cases}$$

Similar to (9), this can be reformulated to a single equation using auxiliary variables s_i and t_i:

$$h(v_i) = \lambda \left(\frac{s_i^2}{2\delta} + t_i \right), \tag{11}$$

where

$$\begin{aligned} s_i + t_i &\geq v_i, \\ 0 \leq s_i &\leq \delta, \\ t_i &\geq 0. \end{aligned}$$

Perhaps unsurprisingly, this epigraph is the mirror opposite of our previously derived reformulation for the Berhu's Loss. (11) can then be made the entire objective function, along with other penalties such as alpha gating, Elastic Net, or Berhu's Loss. In portfolio optimization, minimizing the squared tracking error (deviation from $\tilde{w}$) or optimizing a covariance matrix Σ can make the optimizer sensitive to outliers. If some $\tilde{w}$ predicts an extreme, inaccurate weight for a single stock, the optimizer can appropriate a comparatively massive amount of exposure. A Huber's Loss base objective function applies an L_2 norm for small $|w - \tilde{w}|$, and an L_1 norm for larger turnovers, preventing anomalous $\tilde{w}$ values from hijacking total exposure allocation. The value of λ dictates how much the solver discourages responses to large outliers in $\tilde{w}$.

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